INCH-POUND

MIL-G-45811D(ME)
6 February 1992
SUPERSEDING
MIL-G-45811C(ME)
14 August 1984

MILITARY SPECIFICATION

GAGE, AIR, LOW PRESSURE, PNEUMATIC FLOATING EQUIPMENT

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a direct-reading gage calibrated to read air pressures up to 10 pounds per square inch (psi).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 456 - Electrodeposited Coatings of Copper Plus Nickel Plus Chromium and Nickel Plus Chromium.
ASTM D 2000 - Rubber Products in Automotive Applications.
ASTM D 3953 - Strapping, Flat Steel and Seals.
ASTM D 4675 - Selection and Use of Flat Strapping Materials.
ASTM E 380 - Metric practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The gage shall be as shown in figure 1 and as specified herein. It shall consist of a tubular body containing a graduated indicator bar actuated by air pressure applied through a chuck-type air inlet.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor, shall be compatible with the intended use, and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The gage shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the gage may be newly fabricated from recovered materials to the maximum extent practicable, provided the gage produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the gage.

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3.3.3 Coating and anodizing. Brass and bronze shall be nickel-chromium coated in accordance with ASTM B 456, service condition SC 3, finish optional. Aluminum shall be anodized to a thickness of 0.0007 inch in accordance with MIL-A-8625, type II, class 1.

3.4 Characteristics. The gage shall withstand a pressure of 20 psi applied through the air inlet without cracking, rupture, permanent deformation or other damage and without affecting subsequent accuracy. The gage shall weigh not more than one-half pound. All functioning surfaces of the gage shall be smooth and free from tool marks and extraneous materials. Exterior surfaces shall be free of any sharp edges or projections, and required edges shall be broken with a radius or chamfer, as applicable.

3.5 Body. The body shall be brass, bronze, or aluminum at the contractor's option and shall incorporate a guide for the indicator bar.

3.6 Indicator bar. The indicator bar shall be aluminum or nylon at the contractor's option, and shall be square in cross section. The bar shall be graduated from 1 psi to 10 psi in increments of 1/2 psi on two sides. Pound increments shall be numbered, in black, as shown on figure 1. The other two sides shall be marked in the International system of units (SI) in accordance with ASTM E 380. The SI metric marking shall be in units of kilopascal (kPa). The labels "psi" and "kPa" shall be located on the appropriate sides of the indicator bar near the end protruding from the gage body. The SI marking shall be in red and shall be graduated in 5 kPa increments with numeric marking every 10 kPa reading from 10 to 70 kPa minimum range. Numerals shall be not less than 1/8 inch high. Graduations and numerals shall be on all four sides and shall be clear cut and well defined, providing contrast between the markings and the bar. The pressure readings shall be taken directly from the indicator bar at the plane of emergence from the body. When applicable, the aluminum indicator bar shall include a spring to return the bar to the zero-pressure position when air pressure is removed from the air inlet. The spring shall be corrosion-resistant metal or metal treated to resist corrosion. The use of the return spring is optional with the nylon bar, when applicable.

3.7 Air inlet. The air inlet shall be the chuck-type as shown on figure 1. The metal portion of the chuck shall be the same material as the body. The chuck shall include a cured polychloroprene synthetic rubber chuck foot bonded to the metal. The rubber shall conform to ASTM D 2000, grade M2BC510A14B14C12 and shall be no more than 12 months old on date of acceptance by the Government. The air inlet shall have a screen or other filtering device to prevent foreign particles from entering and damaging the gage.

3.8 Internal mechanism. All parts of the internal mechanism shall be of corrosion-resisting material, and shall be electrolytically compatible with other materials.

3.9 Drop resistance. When subjected to the test specified in 4.5.2.2, the gage shall show no evidence of rupture, cracking or permanent deformation.

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3.10 Accuracy. The gage shall be accurate within 1/2 psi at all pressures from 1 psi to 10 psi throughout a temperature range of 10 °F to 125 °F.

3.11 Identification marking. Each gage shall be identified in accordance with MIL-STD-130 and the marking shall be located as shown on figure 1.

3.12 Workmanship. The gage shall be free of any defects which could impair the operation and accuracy of the gage. The body of the gage shall be smooth and free from sharp edges. The metal portion of the chuck shall be free of sharp edges or protrusions; and the rubber chuck foot shall be smooth and free from mold marks or other defects that could adversely affect performance of the gage. The indicator bar shall not bind in the body.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specification and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The first article gage shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article gage and material representative of that used in the rubber chuck foot shall be tested as specified in 4.5.2 through 4.5.2.5. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.1.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. A lot shall be accepted when 0 defects are found and rejected when one or more defects are found.

4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2 through 4.5.2.4, except that the accuracy test (4.5.2.4) shall be conducted only at 70 °F. A lot shall be accepted when 0 defects are found and rejected when one or more defects are found.

4.5 Inspection procedure.

4.5.1 Examination. The gages shall be examined as specified herein for the following defects:

101. Components not as specified.
102. Dimensions not as specified.
103. Material not as specified.
104. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Contractor does not have documentation available for identification of material, material finishes or treatments.
107. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the gage.
108. Finish not as specified.
109. Weight not as specified.

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110. Graduations and numerals on indicator bar not as specified, missing, or not clear cut and well defined.
111. Identification marking not as specified.
112. Workmanship not as specified.

4.5.2 Tests. The gage shall be subjected to the following tests for over pressure, drop resistance , and endurance in the sequence listed. After completion of each test the gage shall be subjected to the accuracy test specified in 4.5.2.4.

4.5.2.1 Overpressure test. The gage shall be subjected to a pressure (air or nitrogen) of 20 psi applied through the air inlet. Nonconformance to 3.4 and to 3.10 shall constitute failure of this test.

4.5.2.2 Drop resistance test. The gage shall be dropped on its side four times and on each end two times from a height of 4 feet into a hard concrete surface. Nonconformance to 3.9 and to 3.10 shall constitute failure of this test.

4.5.2.3 Endurance test. The gage shall be actuated by the application of air pressure of 5 to 10 psi, and the indicator bar closed after each application of air, for not less than 300 gaging operations or cycles. The gaging operations or cycles shall be conducted at a rate of 20 to 40 per minute. Any damage to the gage, inability to operate or nonconformance to 3.10 shall constitute failure of this test.

4.5.2.4 Accuracy test. The gage shall be subjected to pressure (dry air or dry nitrogen) ranging from 0 to 10 psi. The gage shall be calibrated against a mercury manometer (corrected to standard conditions) connected to the same pressure source. Calibration readings shall be taken from 0 to 10 psi and from 10 psi to 0. Pressure readings shall be recorded at each 1-psi interval. Calibration readings shall be taken at gage temperatures of 70 °F, 10 °F, and 125 °F. The gage shall be conditioned at the test temperature for not less than 1 hour prior to taking readings. Readings of the gage not matching the readings of the mercury manometer within 1/2 psi shall constitute failure of this test.

4.5.2.5 Rubber chuck foot. Nonconformance of the rubber chuck foot to the requirements of 3.7 shall constitute rejection of the air inlet assembly.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. One or more defects shall be cause for rejection.

113.	Preservation not as specified for level A.
114.	Quantities consolidated together not as specified for level A.
115.	Consolidation boxes not closed and sealed as specified for level A.
116.	Preservation and packing not in accordance with the referenced document as specified for level C.
117.	Gross weight exceeds 200 pounds for level A.
118.	Strapping not as specified for level A.
119.	Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each gage shall be preserved in accordance with MIL-P-116, method IC-1, or IC-2 if the gage is furnished in the manufacturer's container.

5.1.1.1 Consolidation. Gages preserved as specified in 5.1.1 shall be consolidated together, in quantities of 20, in a close fitting box conforming to PPP-B-636, W6c. The box shall be closed and secured in accordance with method V of the appendix to the box specification.

5.1.2 Level C. Gages shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Gages, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 200 pounds. Box closure shall be in accordance with the appendix to the box specification. Strapping shall conform to ASTM D 4675 and ASTM D 3953, type 1 or 2, zinc-coated, size as applicable.

5.2.2 Level B. Gages, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-636, V3c, style optional. The gross weight or size of the box shall not exceed the limitations of the box specification. The box shall be closed and secured in accordance with the appendix to the box specification.

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5.2.3 Level C. Gages shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gage is intended for use in measuring air pressure in low pressure (1 to 10 psi) pneumatic equipment such as pontoon bridge floats, bridge erection rollers, reconnaissance boats, and assault boats.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and the number of units required (see 3.2).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (keyword) listing.

Boats
Inflatable
Measurement

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Army - AT

Preparing activity:
Army - ME

Project 5420-A372

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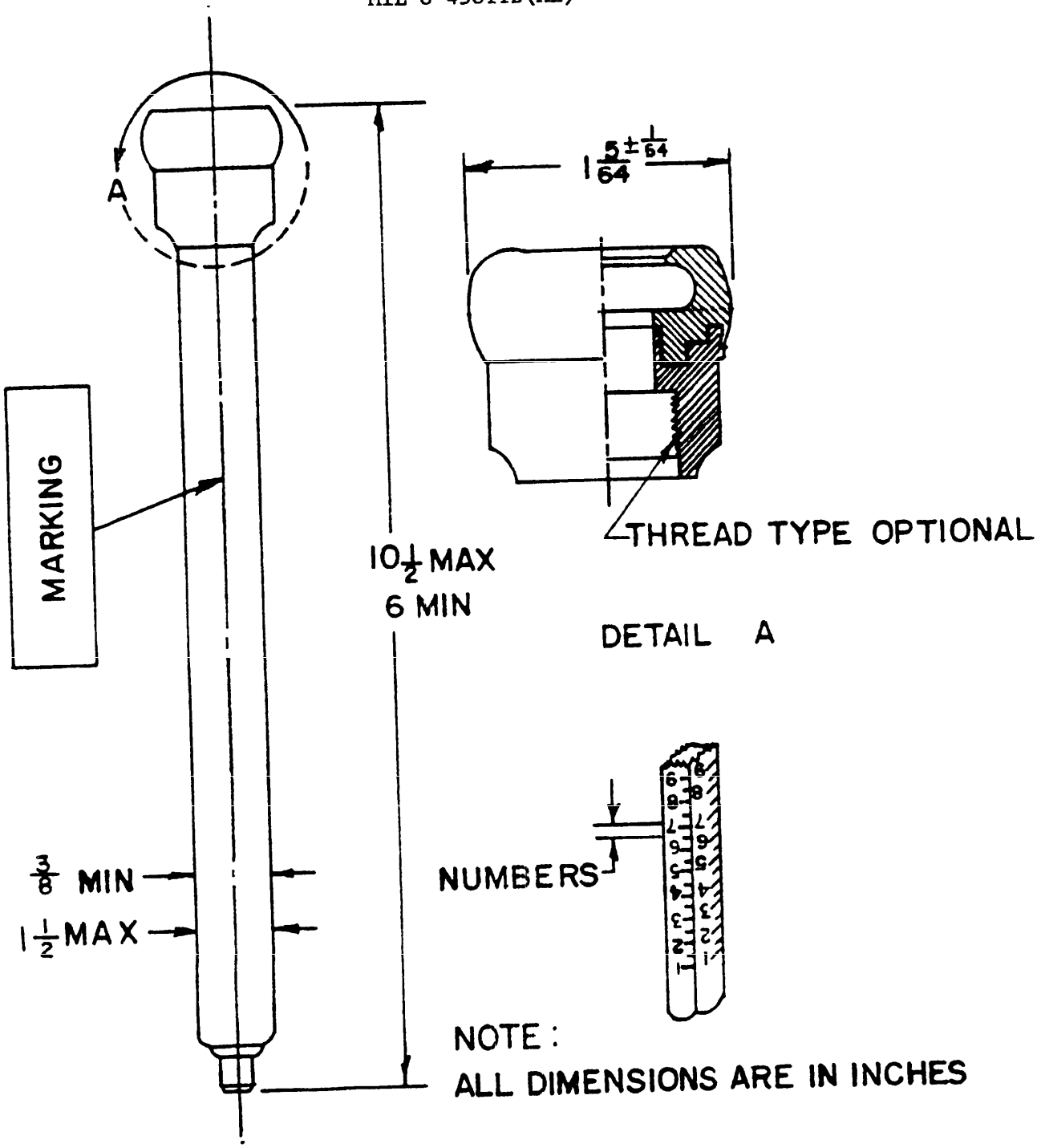


FIGURE 1. Low pressure gage.

X-2463B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-G-45811D(ME)	2. DOCUMENT DATE (YYMMDD) 920206

3. DOCUMENT TITLE: Gage, Air, Low Pressure, Pneumatic Floating Equipment

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Gerry Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806 (2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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